ANALYTICAL CUMULATIVE EXAM

October 4, 1997, Theme: Chemiluminescence and Absorbance

Chemiluminescence

1. Nitric oxide (NO) is a simple molecule that transmits signals between cells in various tissues. It can be measured at the picomolar level by its chemiluminescent reaction with the compound luminol in the presence of H_2O_2.

(a) (15 points) Design an instrument that you could use to measure nitric oxide concentrations. Specifically list any components that you would need. Draw a diagram showing the sample location and identify and locate other components.

(b) (35 points) You perform an experiment, using your new instrument. To a 2.00 mL aliquot of a sample containing an unknown amount of NO, the following was added: 1.00 mL of water, 2.00 mL of a dilute H_2O_2 solution, and 1.00 mL of an alkaline solution of luminol. The chemiluminescent signal from the mixture was integrated over a given period and found to be 21.1.

To a second 2.00 mL aliquot of the sample, the following was added: 1.00 mL of 1.0×10^{-11} M NO, 2.00 mL of a dilute H_2O_2 solution, and 1.00 mL of an alkaline solution of luminol. The integrated intensity was 39.2. Calculate the NO molarity of the sample.

2. (20 points) Chemiluminescence can also occur in reactions involving electochemically-generated species. Electrochemistry is convenient for generating a reactant if it is a radical, unstable, or difficult to store or handle. The following reactions show how chemiluminescence can arise from a reactant ($Ru(bpy)_3^{3+}$, where bpy = bipyridyl) that is electrochemically generated from $Ru(bpy)^{2+}$. X_{red} is the analyte in solution that is to be detected. (The * means that the species is in an excited state.)

$$Ru(bpy)_3^{2+} \rightarrow Ru(bpy)_3^{3+} + 1\,e^- \qquad E^\circ = +1.32 \text{ vs. SCE}$$

$$Ru(bpy)_3^{3+} + X_{red} \rightarrow Ru(bpy)_3^{2+*} + X_{ox}$$

$$Ru(bpy)_3^{2+*} \rightarrow Ru(bpy)_3^{2+} + h\upsilon$$

Design an experimental setup that you could use to measure X_{red} concentration in a sample. Specifically list any components that you would need. Draw a diagram showing the sample location and identify and locate other components. Describe the species in the starting solution and possible electrodes that could be used. What potential would you use to get chemiluminescence to occur? What calibrations or procedures would you use to determine the unknown concentrations of X_{red} from the chemiluminescent signal?

Absorbance: Deviations from Beer's Law

3. For the following situations, 3a, 3b, and 3c, indicate whether i, ii, or iii is the best description for a deviation from linearity, if any, in a plot of absorbance vs. concentration (i.e. deviation from Beer's Law). Give an explanation for each answer.

(i) Plot remains linear

(ii) Plot deviates negatively from linearity (decreased change in absorbance at higher concentrations).

(iii) Plot deviates positively from linearity (increased change in absorbance at higher concentrations).

Indicate whether i, ii, or iii occurs for the following conditions, and give an explanation:

3a. (10 points) The absorber is at a concentration that is very high (much greater than 0.01 M).

3b. (10 points) The absorber undergoes equilibrium with the solvent, forming a product that absorbs more strongly than and at the same wavelength as the starting species.

3c. (10 points) The absorber undergoes equilibrium with the solvent, forming an absorbing species whose spectrum and molar absorptivity are identical to the starting species.